



Advanced Time Series Analysis and Forecasting

Chapter 8: Advanced volatility modelling: realised measures, HAR and multivariate GARCH



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Acronyms used in this material (1/3)

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ ARCH: AutoRegressive Conditional Heteroskedasticity
- ▣ ARMA: AutoRegressive Moving Average (model)
- ▣ BEKK: Baba–Engle–Kraft–Kroner (multivariate GARCH model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ BNHLS: Barndorff–Nielsen–Hansen–Lunde–Shephard
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BV: Bipower Variation
- ▣ CAC: Cotation Assistée en Continu (CAC 40 index, Paris) — the CAC 40 index of the Paris bourse
- ▣ CCC: Constant Conditional Correlation (model)
- ▣ CLT: Central Limit Theorem
- ▣ DCC: Dynamic Conditional Correlation
- ▣ DCC-NL: DCC with Nonlinear shrinkage of the target (Engle–Ledit–Wolf 2019)
- ▣ DM: Diebold–Mariano (test)
- ▣ EGARCH: Exponential GARCH
- ▣ EODHD: EOD Historical Data (market-data provider)
- ▣ ES: Expected Shortfall
- ▣ FDR: False Discovery Rate
- ▣ FRED: Federal Reserve Economic Data
- ▣ FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GARCH-MIDAS: GARCH with a Mixed Data Sampling long-run component (Engle–Ghysels–Sohn 2013)
- ▣ GARCH-X: GARCH with eXogenous variables
- ▣ GJR: Glosten–Jagannathan–Runkle (asymmetric GARCH)

Acronyms used in this material (2/3)

- ▣ GMM: Generalized Method of Moments
- ▣ GMV: Global Minimum-Variance (portfolio)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ HAR-CJ: HAR with Continuous and Jump components
- ▣ HARQ: HAR with realised Quarticity
- ▣ HEAVY: High-frEquency-bAsed VolatilitY (models)
- ▣ IGARCH: Integrated GARCH
- ▣ INDPRO: Industrial production index (FRED series code)
- ▣ IQ: Integrated Quarticity
- ▣ LLM: Large Language Model
- ▣ MAE: Mean Absolute Error
- ▣ MCS: Model Confidence Set
- ▣ MFM: Modelling Financial Markets
- ▣ MIDAS: Mixed DAta Sampling (regression)
- ▣ ML: Machine Learning
- ▣ MSE: Mean Squared Error
- ▣ MSTR: MicroStrategy (Strategy Inc.), NASDAQ ticker
- ▣ OLS: Ordinary Least Squares
- ▣ OPG: Outer Product of the Gradient (scores)
- ▣ PPI: Producer Price Index
- ▣ QLIKE: Quasi-LIKElihood loss
- ▣ QML: Quasi-Maximum Likelihood
- ▣ QMLE: Quasi-Maximum Likelihood Estimator
- ▣ QV: Quadratic Variation
- ▣ RGARCH: Realized GARCH

Acronyms used in this material (3/3)

- ▣ RK: Realised Kernel
- ▣ RMSE: Root Mean Squared Error
- ▣ RQ: Realised Quarticity
- ▣ RV: Realised Variance
- ▣ SV: Stochastic Volatility
- ▣ TQ: Tripower Quarticity
- ▣ TSA: Time Series Analysis
- ▣ TSLA: Tesla Inc., NASDAQ ticker
- ▣ TSRV: Two-Scales Realised Variance
- ▣ US: United States
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)
- ▣ VEC: VEC multivariate GARCH (half-vectorised covariance)
- ▣ VIX: CBOE Volatility Index
- ▣ VR: Variance Ratio
- ▣ WPSFD49207: PPI by commodity: final demand, finished goods (FRED series code)

Today's question and route

- ▣ **Question:** how precisely can we measure, model and forecast a variance that is never observed, and which of our tools still work when the data are noisy, jumpy or high-dimensional?
 - ▶ two routes to volatility: a parametric filter of daily returns (GARCH) and a nonparametric measurement from intraday prices (realised measures); the frontier combines them
- ▣ **Route** of the chapter
 - ▶ quasi-maximum likelihood for GARCH and robust inference; long-run components: component GARCH and GARCH-MIDAS with macroeconomic drivers
 - ▶ realised measures: asymptotics of realised variance, microstructure noise, realised kernels, bipower variation and jump tests
 - ▶ forecasting with realised measures: HAR, HARQ, Realized GARCH, HEAVY; robust loss functions
 - ▶ multivariate: BEKK, DCC and cDCC, the curse of dimensionality and DCC-NL, realised covariance
- ▣ We build on [TSA, Chapter 5](#) (GARCH, GJR, EGARCH, news impact, QLIKE) and [TSA, Chapter 14](#) (DCC basics); MFM, Chapters 5–9 apply these tools to market risk; Seminar 8 comes before this lecture

Learning outcomes

- ▣ State the conditions under which Gaussian QML for GARCH is consistent and asymptotically normal, and compute Bollerslev–Wooldridge standard errors
- ▣ Separate short-run from long-run volatility with component GARCH and GARCH-MIDAS, and judge whether a macroeconomic driver matters
- ▣ Derive the limit theory of realised variance, explain the effect of noise and jumps, and choose among RV, realised kernels and bipower variation
- ▣ Forecast volatility with HAR, HARQ and Realized GARCH, and compare forecasts with losses that are robust to a noisy proxy
- ▣ Estimate DCC and cDCC models, explain why large covariance matrices need shrinkage, and measure correlation from intraday data

Reading, data and tools

- Backbone: [Francq and Zakoïan \(2019\)](#); [Andersen, Bollerslev, Diebold and Labys \(2003\)](#); [Barndorff-Nielsen and Shephard \(2002\)](#); [Barndorff-Nielsen, Hansen, Lunde and Shephard \(2008\)](#); [Corsi \(2009\)](#); [Hansen, Huang and Shek \(2012\)](#); [Engle \(2002\)](#); [Patton \(2011\)](#)
 - ▶ case studies: [Engle, Ghysels and Sohn \(2013\)](#) (GARCH-MIDAS), [Bollerslev, Patton and Quaedvlieg \(2016\)](#) (HARQ), [Hansen, Huang and Shek \(2012\)](#) (Realized GARCH), [Engle, Ledoit and Wolf \(2019\)](#) (DCC-NL)
- Python Quantlets of this chapter: [Quantlets/Ch_08](#)
 - ▶ QML with sandwich standard errors, component GARCH, GARCH-MIDAS, realised kernels, jump tests, HAR/HARQ, Realized GARCH, DCC/cDCC and nonlinear shrinkage written out in `numpy`
- Lecture notebook: [open in Google Colab](#)

Data used in this chapter

Series	Source	Use
Daily realised measures of six equity indices (5-minute RV, BV, realised kernel, open-to-close return), 2000–2022	Heber, Lunde, Shephard and Sheppard (2009) , v0.3	HAR, RGARCH
Bitcoin and Ether: one-minute prices 2018–2026, one-second prices August 2026	Binance public data (data.binance.vision); measures computed by us	noise, jumps, HARQ
S&P 500, DAX, BET, Bitcoin, 14 US stocks and 3 equity ETFs, daily	EODHD	QML, MIDAS, DCC
EUR/RON reference rate	BNR	QML
US industrial production, PPI finished goods (monthly)	FRED (INDPRO, WPSFD49207)	GARCH-MIDAS

- The Oxford-Man library ended in February 2022 (archived copy, cited as its terms require); Binance one-minute data extend the realised measures to 18 September 2026; a calibrated simulation gives the theory a known truth
- RV: realised variance; BV: bipower variation; PPI: producer price index

From the trading floor to the tick



New York Stock Exchange, 1908

Photo: Helen D. Van Eaton, *Leslie's Monthly Magazine* (1908);
public domain; [Wikimedia Commons](#)

- 1982–1986: ARCH and GARCH Engle (1982), Bollerslev (1986): volatility as a filtered conditional variance of daily returns
- 1992–2004: QML theory Bollerslev and Wooldridge (1992), Lee and Hansen (1994), Lumsdaine (1996), Francq and Zakoïan (2004); components Engle and Lee (1999); DCC Engle (2002)
- 1998–2008: realised variance Andersen and Bollerslev (1998), Andersen, Bollerslev, Diebold and Labys (2003), Barndorff-Nielsen and Shephard (2002); bipower variation Barndorff-Nielsen and Shephard (2004); noise and kernels Zhang, Mykland and Aït-Sahalia (2005), Barndorff-Nielsen, Hansen, Lunde and Shephard (2008)
- 2009–2019: HAR Corsi (2009), Realized GARCH Hansen, Huang and Shek (2012), GARCH-MIDAS Engle, Ghysels and Sohn (2013), HARQ Bollerslev, Patton and Quaedvlieg (2016), DCC-NL Engle, Ledoit and Wolf (2019)
- Today: high-frequency data are cheap; the questions are statistical (noise, jumps, dimension, evaluation)

From TSA to this chapter

- Known ([TSA, Chapter 5](#)): GARCH(1,1), GJR, EGARCH, the news impact curve, ML with Gaussian and Student- t errors, QLIKE; ([TSA, Chapter 14](#)): CCC and DCC
- New here
 - ▶ why Gaussian ML still works when returns are not Gaussian, and which standard errors are then valid
 - ▶ volatility with a slow, macro-driven component; volatility measured from intraday prices, with its own sampling error
 - ▶ models that use realised measures; evaluation against a noisy target; covariance matrices with many assets
- Stochastic volatility (a latent log-variance with its own shock): [Chapter 6](#); Markov-switching GARCH: [Chapter 7](#); VaR and ES backtesting: [Chapter 9](#); rough volatility: [Chapter 10](#)

The Gaussian quasi-likelihood

- Model: $\varepsilon_t = \sigma_t(\theta_0)\eta_t$, η_t i.i.d., $\mathbb{E}\eta_t = 0$, $\mathbb{E}\eta_t^2 = 1$, unknown law; $\sigma_t^2(\theta) = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2(\theta)$
- QMLE: $\hat{\theta} = \arg \max_{\theta} \sum_{t=1}^T \ell_t(\theta)$, $\ell_t(\theta) = -\frac{1}{2} (\ln \sigma_t^2(\theta) + \varepsilon_t^2/\sigma_t^2(\theta))$: the Gaussian likelihood used as an estimating equation
 - ▶ score: $s_t(\theta) = \frac{1}{2} \left(\frac{\varepsilon_t^2}{\sigma_t^2} - 1 \right) \frac{1}{\sigma_t^2} \frac{\partial \sigma_t^2}{\partial \theta}$; at θ_0 , $\mathbb{E}(s_t|\mathcal{F}_{t-1}) = 0$ because $\mathbb{E}(\eta_t^2 - 1) = 0$
 - ▶ only the conditional variance must be right; the shape of the law of η_t is never used
- The score is a martingale difference: a CLT for martingales gives asymptotic normality without independence of ε_t
- The recursion starts from an arbitrary σ_1^2 ; the effect of the start vanishes geometrically when $\beta < 1$

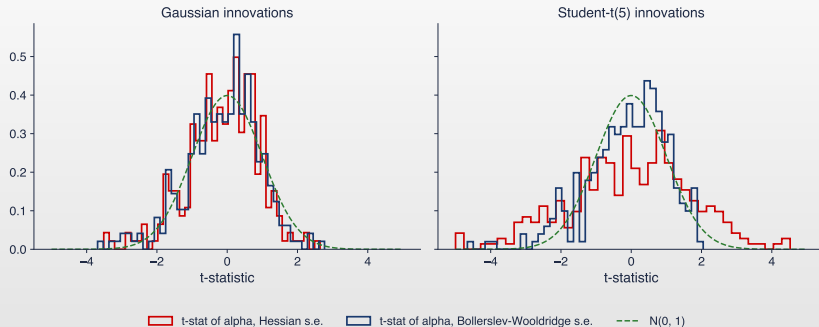
Consistency and asymptotic normality

- Strict stationarity of GARCH(1,1) iff $\mathbb{E} \ln(\alpha_0 \eta_t^2 + \beta_0) < 0$: weaker than $\alpha_0 + \beta_0 < 1$; $\mathbb{E} \varepsilon_t^2$ may be infinite
 - ▶ Lee and Hansen (1994), Lumsdaine (1996): GARCH(1,1), including IGARCH; Berkes, Horváth and Kokoszka (2003), Francq and Zakoïan (2004): GARCH(p, q) under strict stationarity
- Consistency Francq and Zakoïan (2004): strict stationarity, θ_0 in a compact set, identifiability (η_t^2 not degenerate); no moment of ε_t is needed
 - ▶ asymptotic normality adds θ_0 interior and $\kappa_\eta = \mathbb{E} \eta_t^4 < \infty$: $\sqrt{T}(\hat{\theta} - \theta_0) \rightarrow N(0, (\kappa_\eta - 1)J^{-1})$, $J = \mathbb{E} \left[\frac{1}{\sigma_t^4} \frac{\partial \sigma_t^2}{\partial \theta} \frac{\partial \sigma_t^2}{\partial \theta'} \right]$
- Even explosive GARCH ($\mathbb{E} \ln(\alpha_0 \eta_t^2 + \beta_0) > 0$): $(\hat{\alpha}, \hat{\beta})$ remain consistent and asymptotically normal, ω is not identified Jensen and Rahbek (2004)
- Boundary: if $\alpha_0 = 0$ the limit is the projection of a normal vector on a cone, which is why TSA, Chapter 5 tests ARCH effects one-sided

The sandwich and the Bollerslev–Wooldridge standard errors

- General QML result: $\sqrt{T}(\hat{\theta} - \theta_0) \rightarrow N(0, A^{-1}B A^{-1})$, $A = -\mathbb{E} \partial^2 \ell_t / \partial \theta \partial \theta'$, $B = \mathbb{E} s_t s_t'$
- For GARCH: $A = \frac{1}{2}J$ and $B = \frac{\kappa_\eta - 1}{4}J$, hence $A^{-1}B A^{-1} = (\kappa_\eta - 1)J^{-1}$ (▶ Appendix)
 - ▶ Hessian-only s.e. assume $\kappa_\eta = 3$ and give $2J^{-1}$: too small by the factor $\sqrt{(\kappa_\eta - 1)/2}$
 - ▶ outer-product (OPG) s.e. use $B^{-1} = \frac{4}{\kappa_\eta - 1}J^{-1}$: even smaller when $\kappa_\eta > 3$
- Bollerslev and Wooldridge (1992): estimate A by the observed Hessian and B by $\frac{1}{T} \sum \hat{s}_t \hat{s}_t'$; valid whatever the law of η_t (with $\kappa_\eta < \infty$)
- Robust Wald and score tests use the sandwich; the quasi-LR statistic is no longer χ^2 , but a weighted sum of χ_1^2 variables

Hessian against sandwich: a Monte Carlo

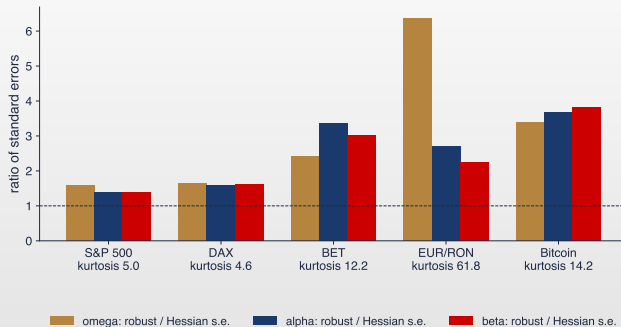


- GARCH(1,1) with $(\omega, \alpha, \beta) = (0.05, 0.08, 0.90)$, $T = 2000$, 300 replications; Gaussian QML in each; histogram of $(\hat{\alpha} - \alpha_0)/\hat{se}$

Interpreting the Monte Carlo

- ▣ Gaussian η : coverage of nominal 95% intervals for α is 95.0% (Hessian) and 94.3% (sandwich): both fine
- ▣ Student- $t(5)$, $\kappa_\eta = 9$: Hessian intervals cover 73.0% for α and 74.7% for β ; sandwich intervals 91.7% and 92.3%
- ▣ The theory predicts the factor $\sqrt{(9-1)/2} = 2$ between the two standard errors; the point estimates are the same
- ▣ With fat tails, a Hessian-based t -test of α rejects far too often: report sandwich standard errors by default

Robust and naive standard errors in five markets



- Gaussian QML of GARCH(1,1), daily returns 2010–2026; bars: sandwich s.e. divided by Hessian s.e.; labels: kurtosis $\hat{\kappa}_\eta$ of the standardised residuals

Interpreting the five markets

- ▣ S&P 500: $\hat{\alpha} = 0.154$ with s.e. 0.013 (Hessian) against 0.018 (sandwich); $\hat{\kappa}_\eta = 5.0$, theoretical factor 1.42
- ▣ BET: $\hat{\kappa}_\eta = 12.2$, the sandwich s.e. of $\hat{\alpha}$ is 3.36 times larger; EUR/RON (managed float): 2.71 times, $\hat{\kappa}_\eta = 61.8$
- ▣ The ratios track $\sqrt{(\hat{\kappa}_\eta - 1)/2}$ only roughly: in the data η_t is not i.i.d., and the sandwich does not need it to be
- ▣ Student- t ML raises the log-likelihood by 124.4 (S&P 500) and 233.0 (BET), but is consistent only if the t shape is right; QML needs only the variance equation

Recap: QML for GARCH

- Gaussian QML is consistent if the variance equation is right, whatever the shape of the innovations
- Strict stationarity, not finite variance, is what the theory needs; IGARCH and even explosive GARCH are covered
- With fat tails, Hessian standard errors are too small by $\sqrt{(\kappa_\eta - 1)/2}$; use Bollerslev–Wooldridge

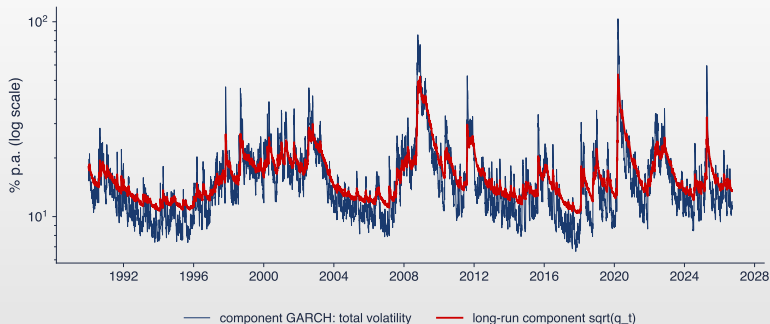
GARCH with exogenous variables

- ▣ GARCH-X: $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 + \pi' x_{t-1}$, with $x_{t-1} \geq 0$ and $\pi \geq 0$ for positivity (or a log specification)
- ▣ Candidates for x_t : a realised measure (leads to HEAVY and Realized GARCH below), the VIX, a macro or policy-uncertainty index, trading volume
 - ▶ QML theory carries over if x_t is stationary and ergodic and its own dynamics need not be modelled for one-step forecasts
 - ▶ multi-step forecasts need a model for x_t : this is exactly what Realized GARCH and HEAVY add
- ▣ A slow variable (a monthly macro series) cannot enter a daily recursion at its own frequency: one needs a component structure (next slides)

The component GARCH of Engle and Lee

- ▣ Engle and Lee (1999): $\sigma_t^2 = q_t + \alpha(\varepsilon_{t-1}^2 - q_{t-1}) + \beta(\sigma_{t-1}^2 - q_{t-1})$, $q_t = \omega + \rho q_{t-1} + \phi(\varepsilon_{t-1}^2 - \sigma_{t-1}^2)$
- ▣ q_t : a slowly moving long-run level (persistence ρ , close to 1); $\sigma_t^2 - q_t$: a transitory deviation (persistence $\alpha + \beta < \rho$)
 - ▶ both are driven by the same shock $\varepsilon_{t-1}^2 - \sigma_{t-1}^2$: the model is a restricted GARCH(2,2)
- ▣ Half-life of a shock to each component: $\ln 0.5 / \ln \rho$ and $\ln 0.5 / \ln(\alpha + \beta)$ days; GARCH(1,1) forces a single half-life
- ▣ Two exponentials approximate a slowly decaying (hyperbolic) autocorrelation of squared returns over a finite range of lags; long memory proper: [Chapter 10](#)

Component GARCH for the S&P 500



□ Daily returns since 1990, $T = 9\,240$; Gaussian QML; annualised total volatility $\sqrt{252\sigma_t^2}$ and long-run component $\sqrt{252q_t}$

Interpreting the component model

- ▣ $\hat{\rho} = 0.995$, $\hat{\phi} = 0.026$; transitory part $\hat{\alpha} + \hat{\beta} = 0.950$: half-lives 150 days (long run) and 13.6 days (short run)
- ▣ GARCH(1,1) on the same data: persistence 0.983, one half-life of 41 days, a compromise between the two
- ▣ Quasi-LR statistic against GARCH(1,1): 44.1; under the null $\phi = 0$ the parameter ρ is not identified (Davies problem), so the χ^2_2 reference is only indicative
- ▣ After 2008, 2020 and 2025 the long-run level stays high for months while daily volatility has already fallen: this matters for horizons beyond a few weeks

GARCH-MIDAS: a macro-driven long-run component



Eric Ghysels, 2019

Photo: Eghysels (2019); CC BY-SA 4.0;

Wikimedia Commons

- Engle, Ghysels and Sohn (2013): day i of month t : $r_{i,t} = \mu + \sqrt{\tau_t g_{i,t}} \eta_{i,t}$
- short run, unit-mean GARCH: $g_{i,t} = (1 - \alpha - \beta) + \alpha \frac{(r_{i-1,t} - \mu)^2}{\tau_t} + \beta g_{i-1,t}$
- long run, MIDAS in a monthly regressor: $\ln \tau_t = m + \theta \sum_{k=1}^K \varphi_k(w) X_{t-k}$
- beta weights restricted to decay: $\varphi_k(w) \propto (1 - k/K)^{w-1}$, $w \geq 1$ Ghysels, Santa-Clara and Valkanov (2006)
- X : realised variance of past months (in levels, $\tau_t = m + \theta \sum \varphi_k RV_{t-k}$) or macro data: industrial production growth, PPI inflation

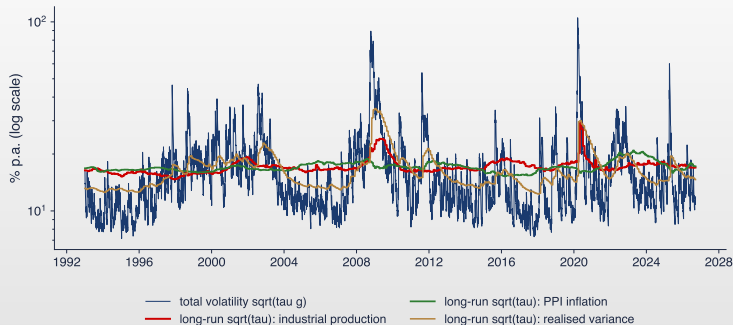
GARCH-MIDAS: identification, estimation and the variance ratio

- ▣ τ_t is constant within the month and $\mathbb{E}g_{i,t} = 1$: the split is identified by the different frequencies, without a latent-variable filter
 - ▶ Gaussian QML of all parameters $(\mu, \alpha, \beta, m, \theta, w)$ in one step; sandwich standard errors
- ▣ Variance ratio $VR = \text{Var}(\ln \tau_t) / \text{Var}(\ln \tau_t g_{i,t})$: the share of the variation of log volatility explained by the long-run component
 - ▶ a macro variable can be significant ($\hat{\theta} \neq 0$) and still explain little of daily volatility (small VR)
- ▣ Inference on w is nonstandard when $\theta = 0$ (not identified) and at the boundary $w = 1$ (flat weights)
- ▣ Extensions: two-sided weights, several regressors, a second (daily) component [Conrad and Kleen \(2020\)](#); forecasts beyond one month use only the long-run part

Case study: Engle, Ghysels and Sohn (2013) on our data

- **Question** of Engle, Ghysels and Sohn (2013): does the stock market's long-run volatility move with macroeconomic fundamentals (output growth, inflation)?
 - ▶ their sample of US stock returns is much longer than ours; their benchmark long-run component is a fixed-window realised variance
- **Our replication:** S&P 500 daily returns, common sample from 04.01.1993, $T = 8482$; $K = 36$ monthly lags; restricted beta weights
 - ▶ three long-run drivers: monthly realised variance (sum of squared daily returns), industrial production growth (FRED INDPRO), PPI inflation (FRED WPSFD49207)
 - ▶ all models against GARCH(1,1) on the same days; QML with sandwich standard errors
- Extension for a project: Romanian industrial production and the BET (Seminar 8, B5); real-time vintages of the macro data

GARCH-MIDAS for the S&P 500



□ Annualised $\sqrt{252\tau_t g_{i,t}}$ of the industrial-production model and the long-run components $\sqrt{252\tau_t}$ of the three models

GARCH-MIDAS estimates

Long-run driver	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\theta}$ (s.e.)	$\hat{w}$	VR (%)	BIC
realised variance (levels)	0.119	0.841	0.0225 (0.0058)	5.939	25.1	22710.0
industrial production growth	0.110	0.870	-0.551 (0.353)	3.692	4.5	22732.5
PPI inflation	0.108	0.873	0.753 (1.260)	1.000	2.2	22733.3
GARCH(1,1)						22721.9

- Same days for all models; $BIC = -2 \ln L + k \ln T$ (lower is better); s.e.: Bollerslev–Wooldridge

Interpreting GARCH-MIDAS

- ▣ Realised variance as the slow driver: $\hat{\theta} = 0.0225$ ($t = 3.86$), $VR = 25.1\%$, BIC below GARCH(1,1): a long-run component exists
- ▣ Industrial production: $\hat{\theta} = -0.551$ ($t = -1.56$), the countercyclical sign of [Engle, Ghysels and Sohn \(2013\)](#), but VR only 4.5% and a higher BIC than GARCH(1,1)
- ▣ PPI inflation: $\hat{w}$ at the boundary 1 (flat weights) and $\hat{\theta}$ imprecise: no evidence in 1993–2026
- ▣ Replication verdict: the qualitative sign survives, the strength does not; macro effects need long samples with deep recessions, which is why the original study uses a much longer history

Recap: Long-run components

- ▣ Volatility has at least two speeds; a single GARCH persistence averages them
- ▣ GARCH-MIDAS lets a monthly variable drive the long-run level; identification comes from the mixed frequencies
- ▣ Report VR and BIC next to $\hat{\theta}$: significance is not importance

Prices as Itô semimartingales

- Log price on day t , $s \in [0, 1]$: $dX_s = \mu_s ds + \sigma_s dW_s + dJ_s$, with σ_s a stochastic spot volatility and J a finite-activity jump process
- Quadratic variation: $QV_t = IV_t + \sum_{s \leq 1} (\Delta J_s)^2$, integrated variance $IV_t = \int_0^1 \sigma_s^2 ds$
 - ▶ without jumps and with σ independent of W : $r_t | IV_t \sim N(\int \mu, IV_t)$: IV is the variance that matters for the daily return
- Realised variance from n intraday returns $r_{t,i} = X_{i/n} - X_{(i-1)/n}$: $RV_t = \sum_{i=1}^n r_{t,i}^2 \rightarrow QV_t$ in probability as $n \rightarrow \infty$
- The drift does not matter in the limit (it is $O(1/n)$ per interval): volatility becomes observable *ex post*, without a model Andersen, Bollerslev, Diebold and Labys (2003), Barndorff-Nielsen and Shephard (2002)

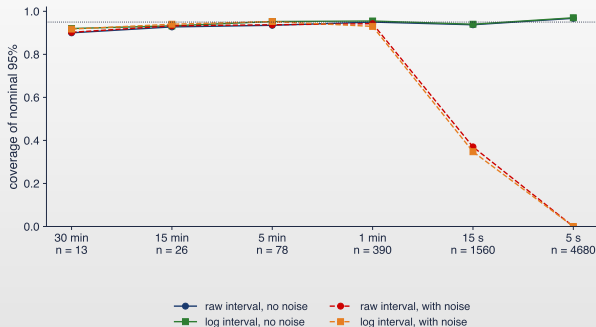
The central limit theorem for realised variance

- Barndorff-Nielsen and Shephard (2002): without jumps, $\sqrt{n}(\text{RV}_t - \text{IV}_t) \rightarrow MN(0, 2\text{IQ}_t)$ stably in law, integrated quarticity $\text{IQ}_t = \int_0^1 \sigma_s^4 ds$
 - mixed normal: normal given the path of σ ; *stable* convergence allows dividing by a random, consistently estimated $\sqrt{\text{IQ}_t}$
- Feasible: $\widehat{\text{IQ}}_t = \frac{n}{3} \sum_i r_{t,i}^4$ (realised quarticity RQ), so $\frac{\text{RV}_t - \text{IV}_t}{\sqrt{\frac{2}{3} \sum_i r_{t,i}^4}} \rightarrow N(0, 1)$
- Log version (delta method): $\ln \text{RV}_t \pm 1.96 \sqrt{\frac{2}{3} \sum_i r_{t,i}^4} / \text{RV}_t$; better in finite samples, never negative
- Rate $\sqrt{n}$: with 5-minute returns on a 6.5-hour day ($n = 78$) the relative error is about $\sqrt{2/78} \approx 16\%$ when volatility is constant within the day
- The measurement error $\text{RV}_t - \text{IV}_t$ is heteroskedastic (it scales with IQ_t): this drives HARQ below

A simulated market with a known truth

- ▣ One-second grid, $n = 23\,400$ seconds a day; $\ln \sigma_s^2 = c + f_s^{(1)} + f_s^{(2)}$, two Gaussian AR(1) factors with half-lives of 60 days and 2 days
 - ▶ stationary s.d. 0.75 and 0.45, mean IV = 1 (%² a day): the mean matches the S&P 500 realised kernel of 2000–2022
 - ▶ leverage: correlation -0.6 between price shocks and shocks to the fast factor
- ▣ Jumps: Poisson with 0.08 jumps a day, sizes $N(0, 0.8^2)$ (%): jumps make about 5.6% of quadratic variation
- ▣ Noise: observed $Y_s = X_s + u_s$, u_s i.i.d. $N(0, \omega^2)$, $\omega = 0.004\%$, so $2n\omega^2 = 0.75$ at one second
- ▣ Each estimator below is compared with the true IV or QV of the same simulated day

Coverage of the feasible CLT



- 400 simulated days without jumps; 95% intervals for IV_t (raw and log) from returns sampled every 30 minutes to every 5 seconds, without and with noise

Interpreting the CLT coverage

- Without noise, coverage approaches 95% as n grows: raw interval 90.0% at $n = 13$ and 95.0% at $n = 390$; the log interval is closer at small n (92.0%)
- With noise, coverage collapses once sampling is too fine: 37.0% at 15 seconds and 0.0% at 5 seconds; at 1 minute it is still 94.2%
- The interval shrinks like $n^{-1/2}$ while the noise bias grows like n : the CLT is a statement about the efficient price, not about the observed one
- Practical rule: 5-minute returns for liquid assets unless a noise-robust estimator is used

Where the realised measures come from



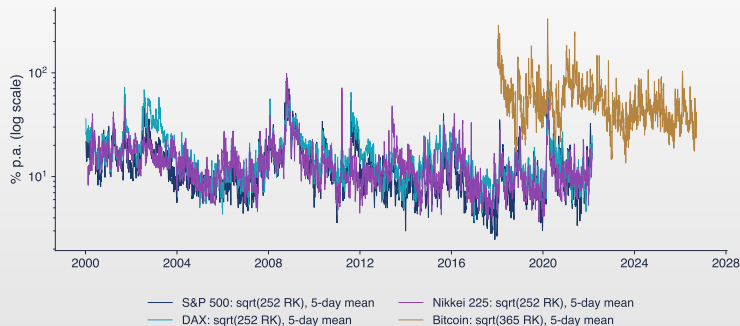
Ole E. Barndorff-Nielsen, 2007

Photo: Thomas Steiner (2007); CC BY-SA 2.5;

[Wikimedia Commons](#)

- [Heber, Lunde, Shephard and Sheppard \(2009\)](#), version 0.3: daily RV, BV, realised kernels and open-to-close returns for 31 indices, 2000 to 25 February 2022, built from cleaned tick data
- its terms: free use if the library and its version are cited; the archived copy keeps the 2022 file available after the shutdown
- Bitcoin and Ether trade 24/7 and Binance publishes all one-minute and one-second prices: we compute RV, BV, RQ, tripower quarticity and a realised kernel for each UTC day, 2018–2026
- No free source covers equity intraday data after 2022: a project can rebuild them from a licensed feed

Realised volatility across markets



- Annualised realised kernel, 5-day means: S&P 500, DAX and Nikkei 225 (Oxford-Man, 2000–2022); Bitcoin (Binance one-minute prices, 2018–2026, 365 days a year)

Interpreting the realised volatilities

- Average annualised volatility from the kernel: S&P 500 16.2%, DAX 19.5%, Nikkei 16.6%, Bitcoin 68.4%
- The equity series move together (2002, 2008, 2011, 2020): a common long-run factor, the multivariate theme of the last section
- Open-to-close measures miss the overnight return: for daily risk one adds the squared overnight return or scales the measure
- Bitcoin has no overnight gap, but its daily volatility is three to four times higher and falls after 2023

Microstructure noise

- Observed log price $Y_{i/n} = X_{i/n} + u_i$: bid–ask bounce, price discreteness, stale quotes; u_i i.i.d. with variance ω^2 , independent of X
- Then $\mathbb{E}(\text{RV}_t^{(n)}|X) = \text{IV}_t + 2n\omega^2$ and $\text{Var}(\text{RV}_t^{(n)}|X) \approx 4n\mathbb{E}u^4$: RV diverges as $n \rightarrow \infty$ (Appendix)
 - ▶ consequence: $\hat{\omega}^2 = \text{RV}_t^{(n)}/(2n)$ at the highest frequency estimates the noise variance
 - ▶ returns of the observed price have first-order autocovariance $-\omega^2$: negative autocorrelation is the signature of i.i.d. noise
- Signature plot: average $\text{RV}^{(n)}$ against the sampling interval; a flat region shows where noise no longer matters
- Real noise is not i.i.d.: it is serially correlated and correlated with the efficient price, especially in quote data Hansen and Lunde (2006b)

Optimal sparse sampling

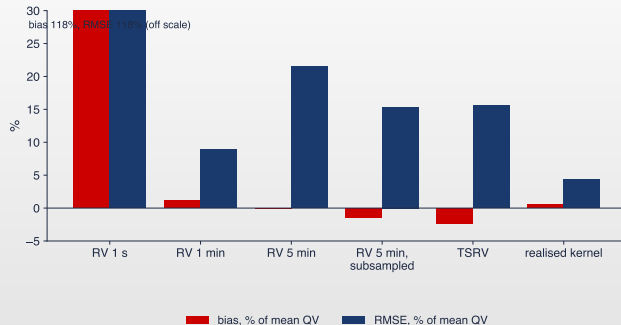
- Mean square error of $RV^{(n)}$ for IV: $MSE(n) \approx \frac{2IQ}{n} + (2n\omega^2)^2 + 4n\mathbb{E}u^4 + \dots$: variance falls, bias rises with n
- Minimising the first two terms: $n^* \approx (IQ/(4\omega^4))^{1/3}$ [Bandi and Russell \(2008\)](#): the optimal grid is coarser when noise is large relative to volatility
 - ▶ example (Seminar 8, A3): $IV = IQ = 1$, $\omega^2 = 1.6 \times 10^{-5}$ gives $n^* \approx 990$, a return every 24 seconds
- The rate of the best sparse RV is only $n^{1/6}$, and most data are thrown away: this motivates estimators that use all observations
- Subsampling: average the RV of the K offset grids (start at second $0, 1, \dots, K-1$); same bias, smaller variance

Two scales and realised kernels

- Two-scales RV [Zhang, Mykland and Aït-Sahalia \(2005\)](#): $\text{TSRV} = \overline{\text{RV}}^{(K)} - \frac{\bar{n}}{n} \text{RV}^{(n)}$, $\bar{n} = (n - K + 1)/K$: the fast scale estimates the noise bias of the slow one; rate $n^{1/6}$
- Realised kernel [Barndorff-Nielsen, Hansen, Lunde and Shephard \(2008\)](#):

$$\text{RK}_t = \gamma_0 + \sum_{h=1}^H k\left(\frac{h}{H+1}\right) (\gamma_h + \gamma_{-h}), \quad \gamma_h = \sum_i r_{t,i} r_{t,i-h}$$
 - ▶ the autocovariances γ_h remove the noise bias (the same idea as HAC variance estimation, [Chapter 0](#)); Parzen weight $k(x) = 1 - 6x^2 + 6x^3$ for $x \leq \frac{1}{2}$, $2(1-x)^3$ for $x > \frac{1}{2}$
 - ▶ nonnegative by construction; rate $n^{1/5}$ with $H = c^* \xi^{4/5} n^{3/5}$, $\xi^2 = \omega^2 / \sqrt{\text{IQ}}$, $c^* = 3.51$ for Parzen [Barndorff-Nielsen et al. \(2009\)](#)
 - ▶ it also corrects serially dependent noise of short range, of either sign
- Pre-averaging [Jacod et al. \(2009\)](#): average returns over blocks before squaring; rate $n^{1/4}$, the optimal one; similar in practice to RK

Estimators against the true quadratic variation

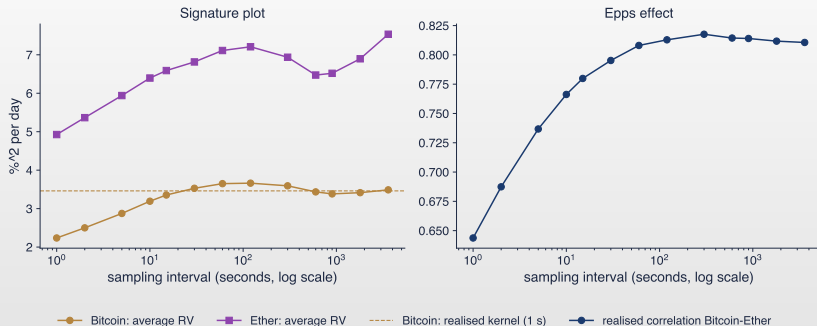


- 300 simulated days with jumps and noise; bias and root mean square error relative to the mean QV; TSRV with $K = 300$ seconds; RK with the BNHLS bandwidth (median $H = 31$)

Interpreting the estimator comparison

- ▣ RV at one second is dominated by noise: bias 117.8% of QV; at one minute the bias is 1.2% and the RMSE 8.9%
- ▣ Five-minute RV is unbiased but noisy (RMSE 21.6%); subsampling and TSRV reduce it to 15.4% and 15.6%
- ▣ The realised kernel uses all 23 400 returns: bias 0.5%, RMSE 4.4%, the best of the six
- ▣ All estimators here target QV, jumps included; separating the jump part needs the next section

Signature plot and Epps effect: Bitcoin and Ether



- Binance one-second prices, August 2026 (31 days); average daily RV by sampling interval (subsampled); realised correlation of the two coins by interval

Interpreting the signature plot

- Bitcoin: RV is 2.24 at one second, 3.65 at one minute, 3.59 at five minutes: the signature *rises*, the opposite of the i.i.d.-noise prediction
- Reason: 52% of one-second returns are zero (no trade or the same price) and prices adjust gradually, so one-second returns are positively autocorrelated
- The realised kernel on one-second data (3.46, median $H = 22$) adds the positive autocovariances back and lands on the flat region of 1–30 minutes
- Epps effect [Epps \(1979\)](#): realised correlation 0.64 at one second against 0.82 at five minutes; asynchronous trading biases covariances towards zero at high frequency

Recap: Realised measures

- ▣ RV estimates QV with error $\sqrt{2IQ/n}$; the log interval behaves better; stable convergence makes studentization valid
- ▣ Noise biases RV at high frequency; its sign and size are empirical: look at the signature plot first
- ▣ Realised kernels use all the data and correct short-range dependent noise; 5-minute RV remains a robust benchmark

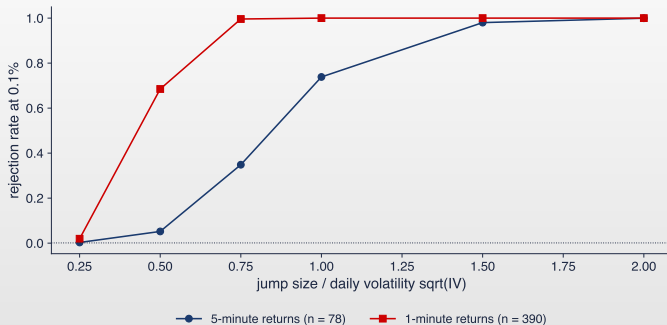
Bipower variation

- [Barndorff-Nielsen and Shephard \(2004\)](#): $BV_t = \mu_1^{-2} \frac{n}{n-1} \sum_{i=2}^n |r_{t,i}| |r_{t,i-1}|$, $\mu_1 = \mathbb{E}|Z| = \sqrt{2/\pi}$; $BV_t \rightarrow IV_t$ even with (finite-activity) jumps
 - a jump enters only two products, each multiplied by a return of order $n^{-1/2}$: its contribution vanishes
- So $RV_t - BV_t \rightarrow \sum (\Delta J)^2$: a nonparametric estimate of the jump variation of the day
- Jump-robust quarticity: tripower $TQ_t = n \mu_{4/3}^{-3} \frac{n}{n-2} \sum_i \prod_{j=0}^2 |r_{t,i-j}|^{4/3}$, $\mu_{4/3} = \mathbb{E}|Z|^{4/3}$
- Nearest-neighbour truncation (MedRV, MinRV) [Andersen, Dobrev and Schaumburg \(2012\)](#): smaller finite-sample bias from a jump and from zero returns
- Threshold (truncated) RV removes returns above $c n^{-\varpi}$, $\varpi \in (0, \frac{1}{2})$: another route to IV

Testing for jumps

- Barndorff-Nielsen and Shephard (2006): without jumps, $\sqrt{n}(\text{RV}_t - \text{BV}_t) \rightarrow MN(0, \vartheta \text{IQ}_t)$,
 $\vartheta = \mu_1^{-4} + 2\mu_1^{-2} - 5 = \frac{\pi^2}{4} + \pi - 5 \approx 0.609$
- Ratio statistic Huang and Tauchen (2005): $z_t = \frac{(\text{RV}_t - \text{BV}_t)/\text{RV}_t}{\sqrt{\frac{\vartheta}{n} \max(1, \text{TQ}_t/\text{BV}_t^2)}} \rightarrow N(0, 1)$; one-sided, reject for large z_t
 - the ratio form and the max adjustment give the best size in their simulations; $\text{TQ}_t/\text{BV}_t^2 \geq 1$ by Jensen when volatility varies within the day
- A daily test answers “was there a jump today?”; locating it within the day needs a test on each return standardised by local BV Lee and Mykland (2008)
- Multiple testing: at level α over T days one expects αT false jump days; use a small α (0.1%) or a family-wise or FDR correction (Chapter 1)
- Separate continuous and jump parts: $C_t = \text{RV}_t - J_t$, $J_t = \mathbb{I}(z_t > z_{1-\alpha})(\text{RV}_t - \text{BV}_t)$ Andersen, Bollerslev and Diebold (2007)

Size and power of the ratio test

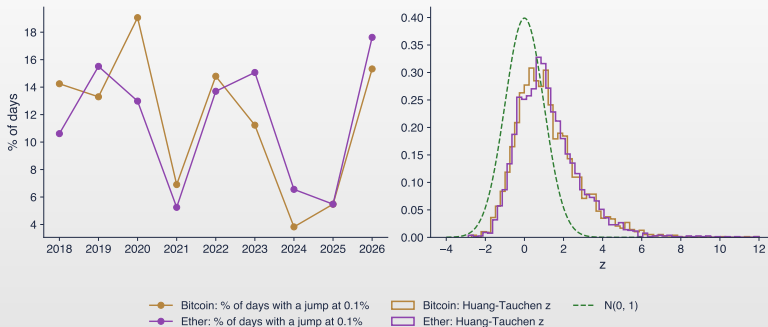


- 2000 simulated days (two-factor SV, no jumps), then one jump of size $c\sqrt{IV_t}$ at a random time; one-sided test at 0.1% with 5-minute and 1-minute returns

Interpreting the size and power

- Size without jumps: 0.0% (5 minutes) and 0.2% (1 minute) against the nominal 0.1%; with i.i.d. noise of the calibrated size still 0.1% at 1 minute
- Power for a jump of half a daily standard deviation: 5% with 5-minute returns, 68% with 1-minute returns; for one standard deviation 74% and 100%
- The power depends on $c\sqrt{n}$: small jumps are absorbed into the diffusion at coarse sampling (Seminar 8, A6 gives the detection threshold)
- The trade-off with noise is the same as for RV: finer sampling raises power until noise and zero returns distort the statistic

Jump days of Bitcoin and Ether



- Huang–Tauchen test with 5-minute returns ($n = 288$ a UTC day), 2018–2026, $T = 3\,165$ days; left: share of jump days at 0.1% by year; right: distribution of z_t against $N(0, 1)$

Interpreting the crypto jump tests

- ▣ Bitcoin: 362 jump days (11.4%) where about 3.2 false rejections are expected; Ether: 355 days (11.2%)
- ▣ Yet the jumps are small: they make 3.0% (Bitcoin) and 2.3% (Ether) of total variation; the rest is continuous
- ▣ The whole distribution of z_t is shifted (mean 1.16, not 0): besides jumps, 24-hour intraday seasonality, zero returns and flash moves violate the assumptions of the null
- ▣ Before calling a day a jump day: standardise returns by an intraday volatility pattern, check zero returns, and control the number of tests

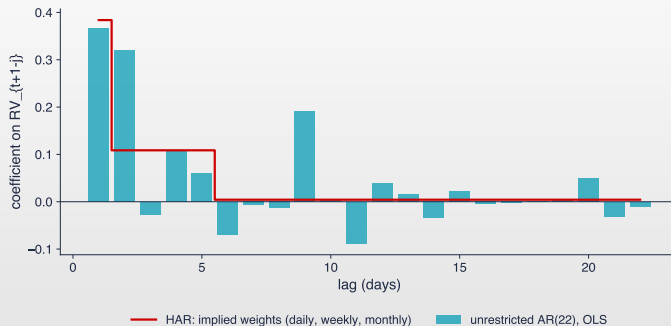
Recap: Jumps

- BV estimates the continuous part; $RV - BV$ the jump part; TQ makes the test robust to jumps in the quarticity
- The ratio test has good size in simulations; its power grows with $c\sqrt{n}$
- In real data the null is fragile: many rejections, little jump variation; treat jump counts as model-dependent

The HAR model of Corsi

- **Corsi (2009):** $RV_{t+1} = \beta_0 + \beta_d RV_t + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + u_{t+1}$, $RV_t^{(w)} = \frac{1}{5} \sum_{j=0}^4 RV_{t-j}$,
 $RV_t^{(m)} = \frac{1}{22} \sum_{j=0}^{21} RV_{t-j}$
- Heterogeneous market hypothesis: daily, weekly and monthly traders react to volatility at their own horizon; the cascade produces slowly decaying autocorrelations
 - ▶ statistically: an AR(22) with 19 linear restrictions (step-shaped coefficients); short memory, but it mimics long memory over a month
- OLS is consistent; the errors are heteroskedastic and, for multi-day targets, overlapping: use Newey–West [Newey and West \(1987\)](#) standard errors
- Variants: log-HAR (on $\ln RV$, near-Gaussian errors; the forecast of RV needs $\exp(\hat{\mu} + \frac{1}{2}\hat{\sigma}_u^2)$), HAR on $\sqrt{RV}$, h -day targets
- Extensions: continuous and jump parts, HAR-CJ [Andersen, Bollerslev and Diebold \(2007\)](#); positive and negative semivariances [Patton and Sheppard \(2015\)](#); implied volatility; leverage terms

HAR as a restricted AR(22)



- S&P 500 (Oxford-Man, 5-minute RV, 2000–2022, $T = 5\,530$): coefficients of an unrestricted AR(22) by OLS and the step weights implied by the HAR estimates

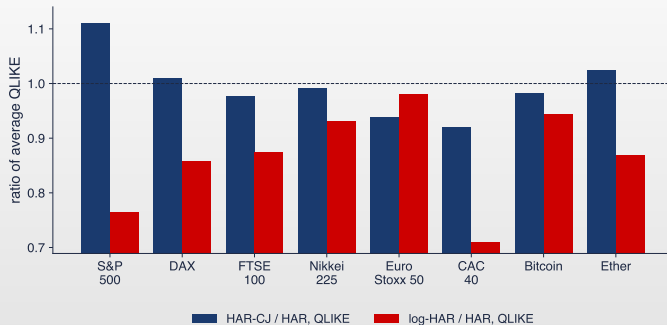
Interpreting the HAR estimates

- ▣ HAR: $\hat{\beta}_d = 0.275$ (0.104), $\hat{\beta}_w = 0.522$ (0.140), $\hat{\beta}_m = 0.094$ (0.078), Newey–West s.e.; $R^2 = 0.560$; persistence $\sum \hat{\beta} = 0.891$
- ▣ The AR(22) with 23 coefficients reaches $R^2 = 0.596$ in sample, with erratic coefficients; HAR captures the shape with 4
- ▣ Log-HAR: $\hat{\beta}_d = 0.393$, $\hat{\beta}_w = 0.399$, $\hat{\beta}_m = 0.153$, $R^2 = 0.719$ on $\ln RV$; in logs the monthly component matters more
- ▣ In levels a few crisis days dominate OLS: large standard errors on β_d and β_m ; weighted least squares or logs give more stable estimates

Forecasting design

- ▣ One day ahead, rolling window of 1 000 days, re-estimated every day; target RV_{t+1} ; losses QLIKE $= \frac{RV}{\hat{f}} - \ln \frac{RV}{\hat{f}} - 1$ and MSE (robust, see below)
- ▣ Insanity filter [Bollerslev, Patton and Quaadvlieg \(2016\)](#): a forecast outside the range of the in-sample RV is replaced by the in-sample mean; it is part of the method and must be reported
 - ▶ linear HAR-type forecasts can be negative after a spike; QLIKE is then undefined
- ▣ Diebold–Mariano [Diebold and Mariano \(1995\)](#) t -statistics of the loss differential against HAR (Newey–West, negative = better than HAR); several models: the model confidence set [Hansen, Lunde and Nason \(2011\)](#), Chapter 1
- ▣ Models: HAR, HAR-CJ (continuous part and jump part, from BV), log-HAR; for Bitcoin and Ether also HARQ

HAR extensions out of sample



- Average QLIKE relative to HAR (below 1 = better), six Oxford-Man indices (2004–2022) and two cryptocurrencies (from 01.11.2020); 1 000-day rolling window

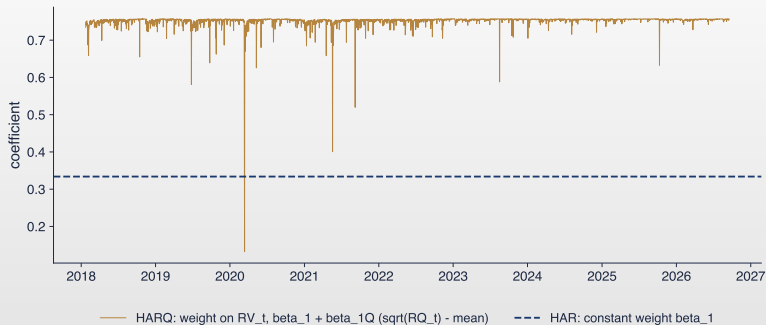
Interpreting the out-of-sample HAR comparison

- Log-HAR beats HAR in QLIKE for 6 of 6 indices: S&P 500 0.764 (DM -2.07), CAC 40 0.709 (DM -1.73), FTSE 0.874 (DM -5.08)
- HAR-CJ helps in 4 of 6 indices, but not for the S&P 500 (1.110): separating jumps adds estimation noise when jumps are small
- The ranking depends on the loss: in MSE the gains of log-HAR are smaller (S&P 500: 0.837); MSE weighs the crisis days most
- Six indices from one data vendor and one period: report all of them, not the best one (data snooping, [Chapter 1](#))

HARQ: exploiting the measurement error

- $RV_t = IV_t + e_t$ with $\text{Var}(e_t|IQ_t) \approx 2IQ_t/n$: the regressor of HAR is measured with an error whose variance changes every day
- Errors in variables: in an AR(1) for IV with coefficient ϕ , the OLS slope on RV tends to $\phi\lambda$, $\lambda = \text{Var}(IV)/(\text{Var}(IV) + \mathbb{E}\text{Var}(e))$ ([▶ Appendix](#))
 - ▶ one constant β_d is a compromise: too high on noisy days, too low on precise days
- **Bollerslev, Patton and Quaedvlieg (2016)**: $RV_{t+1} = \beta_0 + (\beta_d + \beta_{dQ}\sqrt{RQ_t})RV_t + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + u_{t+1}$, expected $\beta_{dQ} < 0$
- One extra parameter, still OLS; $\sqrt{RQ_t}$ demeaned so that β_d is the weight on a day of average precision
- The original paper: S&P 500 futures and 27 Dow Jones stocks, gains over HAR in and out of sample; our replication extends it to Bitcoin and Ether, 2018–2026, where RQ is available from our one-minute data

HARQ for Bitcoin: a weight that moves with precision



- Full-sample HARQ on Bitcoin 5-minute RV and RQ: daily weight $\hat{\beta}_d + \hat{\beta}_{dQ}(\sqrt{RQ_t} - \sqrt{RQ})$ against the constant HAR weight

Interpreting HARQ

- ▣ $\hat{\beta}_{dQ} = -0.00021$ ($t = -2.7$, Newey–West): negative, as predicted; the daily weight at average precision is 0.752, against 0.334 in HAR
- ▣ On most days the weight is near 0.76; on the noisiest days it falls (1% quantile 0.71, minimum 0.13): yesterday's RV is trusted only when it is precise
- ▣ Out of sample, QLIKE relative to HAR: Bitcoin 0.946 (DM -2.44), Ether 0.921 (DM -4.50): the gain of the original study carries over to crypto
- ▣ In MSE the picture differs (Bitcoin 1.163): MSE is dominated by a few extreme days, where HARQ shrinks the most; the AI mini-case checks how fragile the gain is

Recap: HAR and HARQ

- ▣ HAR is a parsimonious restricted AR(22) estimated by OLS; logs stabilise it
- ▣ Out-of-sample gains are loss- and market-specific: report all series, the filter and DM tests
- ▣ HARQ lets the measurement error of RV set the weight of yesterday: one parameter, a robust gain in QLIKE

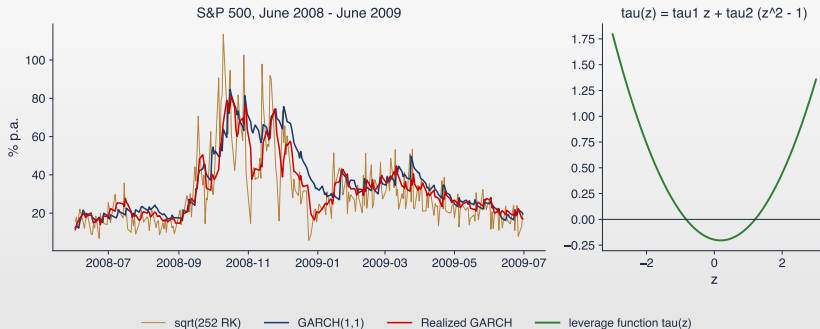
Realized GARCH

- Hansen, Huang and Shek (2012), log-linear form: $r_t = \sqrt{h_t} z_t$; $\ln h_t = \omega + \beta \ln h_{t-1} + \gamma \ln x_{t-1}$;
 $\ln x_t = \xi + \varphi \ln h_t + \tau(z_t) + u_t$
- Measurement equation: the realised measure x_t (RK) is a noisy, possibly biased signal of h_t ; $\varphi = 1$ means proportional, ξ absorbs scale (open-to-close against close-to-close)
 - ▶ leverage function $\tau(z) = \tau_1 z + \tau_2(z^2 - 1)$: negative returns raise the next realised measure (news impact, TSA, Chapter 5)
- Reduced form: $\ln h_t$ is an AR(1) with persistence $\pi = \beta + \varphi\gamma$; multi-step forecasts follow, because x_t has its own equation
- Joint QML: $\ell = -\frac{1}{2} \sum_t [\ln h_t + z_t^2 + \ln \sigma_u^2 + u_t^2 / \sigma_u^2]$; compare with GARCH only through the partial likelihood of r_t

HEAVY and the family of realised-measure models

- ▣ HEAVY Shephard and Sheppard (2010): $\text{Var}(r_t|\mathcal{F}_{t-1}) = h_t = \omega + \alpha \text{RM}_{t-1} + \beta h_{t-1}$ and $\mathbb{E}(\text{RM}_t|\mathcal{F}_{t-1}) = \mu_t = \omega_R + \alpha_R \text{RM}_{t-1} + \beta_R \mu_{t-1}$
- ▣ Two separate equations, each estimated by QML; momentum: after a shock, h_t keeps rising while μ_t catches up
 - ▶ when the realised measure is in the variance equation, the squared return usually adds nothing (ε_{t-1}^2 gets a zero coefficient)
- ▣ Realized EGARCH Hansen and Huang (2016): several realised measures and a richer leverage; score-driven versions exist for fat tails
- ▣ Compared with HAR: returns and realised measures are modelled jointly, so densities and VaR/ES forecasts are available (Chapter 9)

Case study: Realized GARCH for the S&P 500



- Open-to-close returns and the Parzen realised kernel (Oxford-Man), 2000–2022, $T = 5552$, as in Hansen, Huang and Shek (2012); left: the 2008 crisis; right: the estimated leverage function

Realized GARCH estimates

ω	β	γ	ξ	φ	τ_1	τ_2	σ_u
0.119 (0.010)	0.678 (0.018)	0.287 (0.016)	-0.456 (0.021)	1.012 (0.026)	-0.073 (0.008)	0.197 (0.010)	0.671 (0.008)

- Bollerslev–Wooldridge s.e. in brackets; persistence $\hat{\pi} = \hat{\beta} + \hat{\varphi}\hat{\gamma} = 0.968$ (GARCH(1,1) on the same returns: 0.989)
- Partial log-likelihood of the returns: Realized GARCH -6899.7, HEAVY -6902.8, GJR -6998.8, GARCH(1,1) -7099.0
- HEAVY variance equation: $\hat{\alpha} = 0.377$ on RK_{t-1} , $\hat{\beta} = 0.691$

Interpreting Realized GARCH

- ▣ $\hat{\phi} = 1.012$: the kernel is proportional to the conditional variance; $\hat{\xi} = -0.456$ with $\hat{\sigma}_u = 0.671$: on average the kernel lies below the conditional variance of the open-to-close return
- ▣ $\hat{\gamma} = 0.287$ is large and $\hat{\beta} = 0.678$ smaller than in GARCH: the realised measure brings information faster than squared returns (left panel, autumn 2008)
- ▣ Leverage: $\hat{\tau}_1 = -0.073$, $\hat{\tau}_2 = 0.197$: an asymmetric parabola, larger for negative z
- ▣ The gain in the partial likelihood over GARCH(1,1) (199.3 points) has the direction of the original paper; HEAVY is almost as good with three parameters

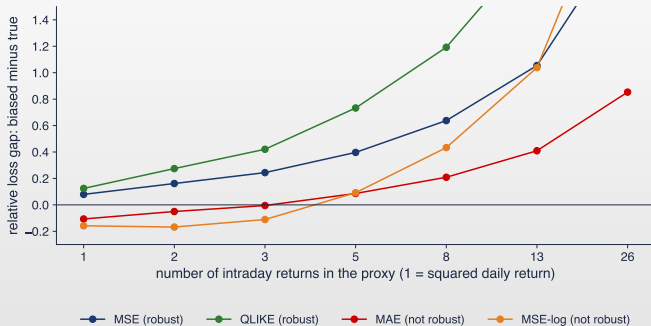
Recap: Realised-measure GARCH

- ▣ A realised measure in the variance equation beats squared returns as the news variable
- ▣ Realized GARCH adds a measurement equation: multi-step forecasts, leverage and a joint likelihood
- ▣ Compare with GARCH only on the partial likelihood of returns

The proxy problem and robust losses

- We never see σ_t^2 ; we compare a forecast h_t with a proxy $\hat{\sigma}_t^2$ (squared return, RV, RK) with $\mathbb{E}(\hat{\sigma}_t^2 | \mathcal{F}_{t-1}) = \sigma_t^2$
- A loss is *robust* [Patton \(2011\)](#) if the ranking of forecasts by $\mathbb{E}L(\hat{\sigma}_t^2, h_t)$ equals the ranking by $\mathbb{E}L(\sigma_t^2, h_t)$ for every unbiased proxy [Hansen and Lunde \(2006a\)](#)
 - ▶ necessary and sufficient: $L(\hat{\sigma}^2, h) = \tilde{C}(h) + B(\hat{\sigma}^2) + C(h)(\hat{\sigma}^2 - h)$, $C' < 0$ (a Bregman form: [Chapter 1](#), consistent scoring functions for the mean)
- Robust: $\text{MSE} = (\hat{\sigma}^2 - h)^2$ and $\text{QLIKE} = \hat{\sigma}^2/h - \ln(\hat{\sigma}^2/h) - 1$ (the members of degree 2 and degree 0 of the homogeneous robust family); QLIKE penalises under-prediction more
- Not robust: MAE, MSE on logs, MSE on standard deviations: with a noisy proxy they reward forecasts that are biased downwards
- Example: with $\hat{\sigma}^2 = r^2$ the MSE-log optimum is $h^* = \exp(\mathbb{E} \ln r^2) = 0.28 \sigma^2$, since $\mathbb{E} \ln \chi_1^2 = -1.27$

Robust and non-robust losses

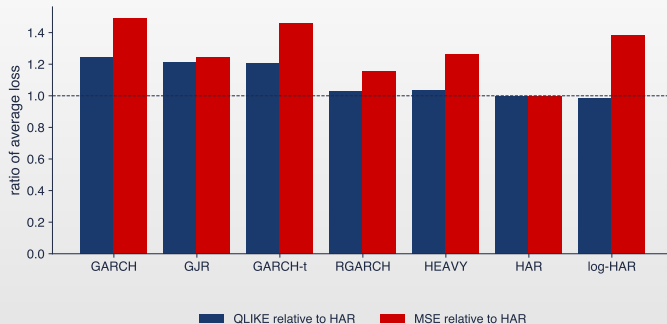


- True variance against a forecast biased down by the factor 0.6; proxy = RV from n intraday returns ($n = 1$: squared daily return); relative gap of expected losses, above zero = the true variance wins

Interpreting the robustness experiment

- ▣ With the squared return as proxy, MAE prefers the biased forecast (gap -10.7%) and so does MSE-log (-15.8%); MSE (7.9%) and QLIKE (12.5%) do not
- ▣ With $n = 5$ intraday returns all four losses rank correctly: a precise proxy makes non-robust losses “almost robust efHLb
- ▣ The danger is largest exactly where realised measures are unavailable: daily data, emerging markets, long histories
- ▣ Rule: evaluate volatility forecasts with QLIKE (and MSE), against the best available proxy

Seven forecasts of S&P 500 volatility



- Open-to-close variance, one day ahead, 2016–2022 (1 537 days); expanding window from 2000, parameters re-estimated every 250 days; proxy: realised kernel

Interpreting the forecast comparison

- ▣ QLIKE: HAR 0.328, log-HAR 0.324, Realized GARCH 0.337, HEAVY 0.339, GARCH- t 0.396, GJR 0.398, GARCH 0.407
- ▣ DM against HAR (QLIKE): GARCH 8.62, GJR 6.27, Realized GARCH 1.42, log-HAR -0.37 : realised-measure models beat return-only models clearly; among them the differences are small
- ▣ In MSE, HAR (1.60) leads and the DM statistics are smaller: the 2020 crash dominates the squared errors
- ▣ One proxy (RK) for all models: the HAR family is fitted to that same measure, which favours it; a fair comparison also tries another proxy (RV, BV)

Recap: Evaluation

- Only losses of the Patton class keep the ranking under a noisy unbiased proxy: MSE and QLIKE
- Realised-measure models beat return-only GARCH one day ahead; DM and the MCS of [Chapter 1](#) quantify by how much
- State the proxy, the loss, the window and the filter: each can change the ranking

From one variance to a covariance matrix

- ▣ $\varepsilon_t = H_t^{1/2} \eta_t$, η_t i.i.d. $(0, I_N)$; H_t must be symmetric positive definite for every t and every parameter value
- ▣ VEC: $\text{vech}(H_t) = c + A \text{vech}(\varepsilon_{t-1} \varepsilon'_{t-1}) + B \text{vech}(H_{t-1})$: general, but positivity is hard to impose
- ▣ BEKK Engle and Kroner (1995): $H_t = CC' + A' \varepsilon_{t-1} \varepsilon'_{t-1} A + B' H_{t-1} B$: positive definite by construction; diagonal and scalar versions restrict A , B
 - identification: (A, B) and $(-A, -B)$ give the same H_t ; fix the sign of one element
- ▣ CCC Bollerslev (1990): $H_t = D_t R D_t$, $D_t = \text{diag}(\sigma_{1t}, \dots, \sigma_{Nt})$ from univariate GARCH; DCC Engle (2002) lets R vary: $H_t = D_t R_t D_t$

The curse of dimensionality in numbers

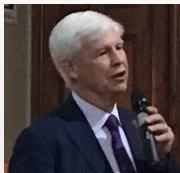
N	VEC	BEKK	diagonal BEKK	scalar BEKK	DCC
2	21	11	7	5	9
5	465	65	25	17	27
10	6 105	255	75	57	77
25	211 575	1 575	375	327	377
50	3 252 525	6 275	1 375	1 277	1 377
100	51 010 050	25 050	5 250	5 052	5 252

- Counts of free parameters of the (1,1) versions; DCC: $3N$ univariate GARCH parameters, 2 correlation parameters and the $N(N-1)/2$ elements of the target
- The DCC target is estimated by moments (correlation targeting), so the numerical search has only 2 parameters at any N ; the price: $N(N-1)/2$ estimated correlations

Asymptotic theory for multivariate GARCH

- Gaussian QML is the standard estimator; consistency and asymptotic normality: VEC and BEKK [Comte and Lieberman \(2003\)](#), [Hafner and Preminger \(2009\)](#); CCC with ARMA means [Ling and McAleer \(2003\)](#)
- Conditions are stronger than in the univariate case (moments of order 6 or 8 for BEKK in early results); the sandwich of the univariate section carries over
- DCC is estimated in two steps [Engle and Sheppard \(2001\)](#): (1) univariate GARCH for each series; (2) QML of the correlation part given the standardised residuals $z_t = D_t^{-1} \varepsilon_t$
 - ▶ second-step standard errors must account for the first step (a GMM-type correction); naive second-step s.e. are too small
- Large N : the full likelihood needs R_t^{-1} and $|R_t|$ at each t ($O(N^3)$) and the score is dominated by noise; composite likelihood over pairs [Pakel, Shephard, Sheppard and Engle \(2021\)](#) avoids both

DCC and its corrected version cDCC



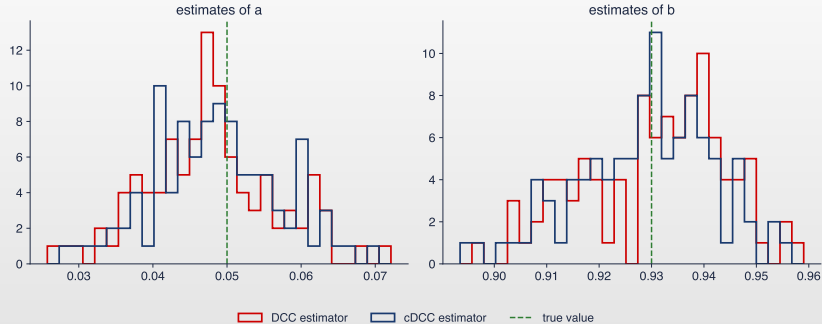
Robert F. Engle, 2017

Photo: Econterms (2017); CC BY-SA 4.0;

[Wikimedia Commons](#)

- DCC: $Q_t = (1 - a - b)S + az_{t-1}z'_{t-1} + bQ_{t-1}$,
 $R_t = \text{diag}(Q_t)^{-1/2}Q_t\text{diag}(Q_t)^{-1/2}$, S = sample correlation of z_t
- Aielli (2013): $\mathbb{E}(z_t z'_t | \mathcal{F}_{t-1}) = R_t \neq Q_t$, so $\mathbb{E}Q_t \neq \mathbb{E}z_t z'_t$ in general: the moment estimator of S is inconsistent, and so is $(\hat{a}, \hat{b})$
- cDCC: replace z_{t-1} by $z_{t-1}^* = \text{diag}(Q_{t-1})^{1/2}z_{t-1}$; then $\mathbb{E}(z_t^* z_t^{*'} | \mathcal{F}_{t-1}) = Q_t$ and $S = \mathbb{E}z_t^* z_t^{*'}$ is a valid target
- the cDCC target depends on (a, b) : estimate it iteratively inside the likelihood

DCC against cDCC: a simulation



- Bivariate cDCC process with $a = 0.05$, $b = 0.93$, target correlation 0.5, unit variances; $T = 2\,000$, 100 replications; both estimators on each sample

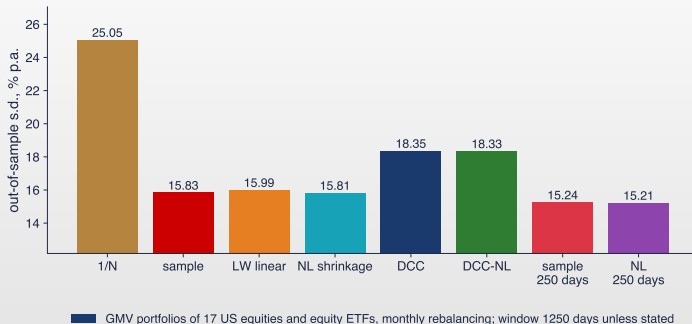
Interpreting the DCC simulation

- Mean estimates: DCC $\hat{a} = 0.0478$, $\hat{b} = 0.9307$; cDCC $\hat{a} = 0.0486$, $\hat{b} = 0.9288$ (true 0.05 and 0.93)
- RMSE of $\hat{a}$: 0.0090 (DCC) and 0.0086 (cDCC); of $\hat{b}$: 0.0135 and 0.0129: the inconsistency is real but small at these persistence levels
- Sampling error dominates the bias with two thousand observations; the difference matters more for the target in large systems and for inference on (a, b)
- A proof of inconsistency does not tell you the size of the bias: simulate it in your own design before choosing an estimator

Large covariance matrices: shrinkage and DCC-NL

- With N/T not small, the eigenvalues of a sample covariance are too dispersed: the largest too large, the smallest too small; its inverse amplifies the error
 - ▶ a GMV portfolio $w = \Sigma^{-1}1/1'\Sigma^{-1}1$ loads on the smallest, most underestimated eigenvalues
- Linear shrinkage [Ledoit and Wolf \(2004\)](#): $\hat{\Sigma} = \delta\mu I + (1 - \delta)S$, δ estimated; nonlinear shrinkage [Ledoit and Wolf \(2020\)](#): keep the eigenvectors of S , replace each eigenvalue λ_i by $d(\lambda_i)$, an analytical kernel formula
- DCC-NL [Engle, Ledoit and Wolf \(2019\)](#): in DCC, the target S of the standardised residuals is itself a large sample covariance: replace it by its nonlinear shrinkage
 - ▶ composite likelihood for (a, b) , univariate GARCH for the variances; evaluated by the out-of-sample s.d. of GMV portfolios
- Why GMV: it depends only on Σ , so its realised risk measures the quality of the covariance forecast without noise from expected returns

Case study: Engle, Ledoit and Wolf (2019) on US equities



- 17 assets: 14 US stocks and three equity ETFs (SPY, QQQ, RSP), which are portfolios of stocks; GMV weights rebalanced every 21 days from 23.05.2017 (112 rebalancings); window 1 250 days, and 250 days for the last two bars; annualised out-of-sample s.d.

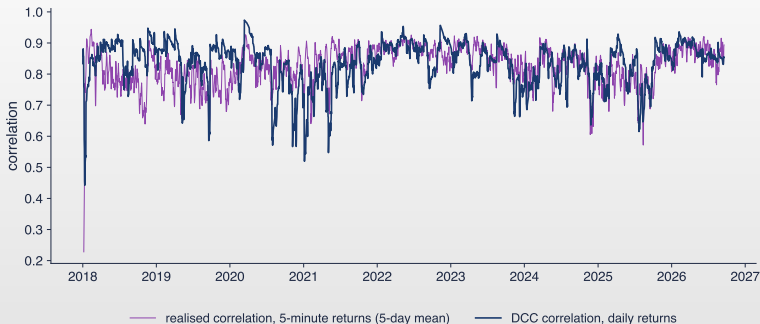
Interpreting the GMV comparison

- Out-of-sample s.d. (% p.a.): 1/ N 25.05, sample 15.83, linear shrinkage 15.99, nonlinear shrinkage 15.81, DCC 18.35, DCC-NL 18.33
- With $N/T = 17/1250$ the sample covariance is already accurate: nonlinear shrinkage changes little (15.81 against 15.83); DCC-NL is no better than DCC
- Here the dynamics hurt: DCC (with GARCH variances, forecasts averaged over the 21-day holding period) gives 18.35 against 15.83; one-month-ahead variance forecasts of single stocks such as MSTR and TSLA are noisy, and GMV amplifies their errors
- With a window of 250 days (N/T five times larger) the sample GMV has s.d. 15.24 and nonlinear shrinkage 15.21: only 0.2% lower, although the ETFs make the correlation matrix nearly singular (median condition number 1146); the shorter window itself lowers the risk, a crude form of time variation
- The original study uses hundreds of stocks, where N/T is large and DCC-NL gains are substantial; with our N the lesson is when shrinkage matters, not how much it gains

Realised covariance

- ▣ $\text{RCov}_t = \sum_i r_{t,i} r'_{t,i} \rightarrow \int_0^1 \Sigma_s ds$ (plus co-jumps); with synchronous, noise-free prices it inherits the CLT of RV
- ▣ Asynchronous trading: previous-tick prices create zero returns for the asset that has not traded; covariances shrink towards zero as the interval falls (the Epps effect)
 - ▶ refresh-time sampling (all assets have traded) and the multivariate realised kernel [Barndorff-Nielsen et al. \(2011\)](#): consistent and positive semi-definite
- ▣ Forecasting: HAR on the elements of the Cholesky factor of RCov_t keeps forecasts positive definite [Chiriac and Voev \(2011\)](#); HEAVY and Realized GARCH have multivariate versions
- ▣ Evaluating covariance forecasts: matrix QLIKE $\ln |H_t| + \text{tr}(H_t^{-1} \Sigma_t)$ and the Frobenius MSE are robust to noisy matrix proxies [Laurent, Rombouts and Violante \(2012\)](#)

Bitcoin and Ether: realised and DCC correlation



- Daily realised correlation from 5-minute returns (5-day means) and the DCC correlation of daily close-to-close (UTC) returns, 2018–2026

Interpreting the correlations

- Average realised correlation 0.83, average DCC correlation 0.83; correlation between the two series 0.26
- DCC uses one observation a day ($\hat{a} = 0.051$, $\hat{b} = 0.928$): it smooths and reacts with a lag; the realised measure moves daily and drops quickly in idiosyncratic episodes (5% quantile 0.68)
- Combining both (a realised measure in the correlation equation) is the multivariate analogue of Realized GARCH
- Five-minute sampling sits on the flat part of the Epps curve for these two coins (signature slide): the realised correlation is not biased towards zero

Recap: Multivariate

- BEKK guarantees positivity at a quadratic parameter cost; DCC separates variances and correlations and scales to large N
- cDCC fixes the inconsistency of the DCC target; in moderate designs the difference is small
- In high dimension, the target needs shrinkage (DCC-NL); evaluate with GMV risk or robust matrix losses

An open question

- Does exploiting the measurement error of realised variance (HARQ) improve volatility forecasts robustly, or only under particular design choices?
 - ▶ formal: the QLIKE ratio HARQ/HAR and its DM statistic across a pre-registered grid (window, filter, realised measure, asset); falsified if the ratio exceeds 1 or the DM test is insignificant for most cells
 - ▶ the measure of precision (RQ) is itself noisy and fat-tailed, so the gain may be fragile
- Why it matters: risk systems use one-day volatility forecasts every day; a small but robust gain is worth more than a large fragile one
 - ▶ literature to start from: [Bollerslev, Patton and Quaedvlieg \(2016\)](#), [Corsi \(2009\)](#), [Patton \(2011\)](#), [Hansen and Lunde \(2006a\)](#), [Hansen, Lunde and Nason \(2011\)](#)

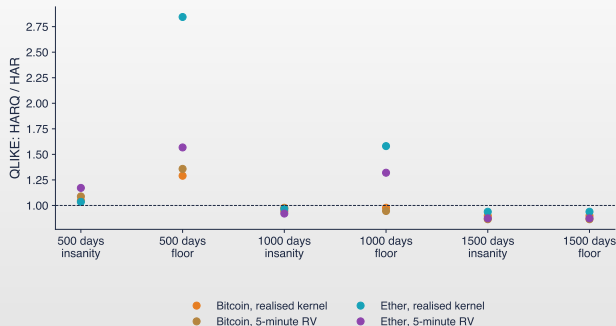
The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
 - ▶ **literature:** List peer-reviewed papers since 2016 that test HARQ-type models out of sample on assets other than US equities; give DOIs. Then check every DOI on Crossref
 - ▶ **hypothesis:** Under which data features (24/7 trading, jumps, zero returns) should the HARQ gain shrink or grow? Give testable predictions.
 - ▶ **code and replication:** ask for HAR and HARQ rolling forecasts, then reproduce a known number first (the Bitcoin QLIKE ratio of this lecture)
 - ▶ **robustness and critique:** Act as a hostile referee: which choices in a HARQ forecast comparison could produce a spurious gain?
- Report: what was asked, what was kept, what was rejected (AI_USE.md, AI_ERRORS.md)

What the human checks

- ▣ Every reference exists and says what is claimed (DOI resolves, title matches, the result is in the paper)
- ▣ The forecast at t uses only data up to t : windows, demeaning of RQ, the insanity filter and the bias correction of log forecasts are all computed in the estimation window
- ▣ The loss is robust (QLIKE, MSE) and the proxy is stated; DM statistics use HAC variances
- ▣ All cells of the grid are reported, not the best one; the number of comparisons is stated
- ▣ Realised measures are recomputed from raw prices and agree with an independent source on overlapping assets and dates

Mini-case: twelve choices, one conclusion?



- QLIKE ratio HARQ/HAR for Bitcoin and Ether: window 500, 1 000 or 1 500 days; insanity filter or a floor at the smallest in-sample RV; target 5-minute RV or realised kernel
- Ratios from 0.865 to 2.844; HARQ better in 58% of the 24 cells, significantly better ($DM < -1.96$) in 50%, significantly worse in 12%: an AI summary reporting one cell would hide most of this

Project idea

- ▣ **Realised volatility of crypto-assets: does the equity toolkit transfer?:** replicate first, then extend
 - ▶ replicate: HAR and HARQ of this lecture for Bitcoin (QLIKE ratio 0.946) and Realized GARCH for the S&P 500 (persistence 0.968)
 - ▶ extend: ten coins from the Binance archive; noise-robust measures from one-second data; an intraday seasonality correction before jump tests; Realized GARCH with the realised kernel; the spot-ETF date (January 2024) as a natural break
 - ▶ pre-register: assets, sample, measures, models, windows, losses, the DM/MCS tests and the robustness grid
- ▣ Deliverables follow the course rules: repository, report, AI_USE.md, AI_ERRORS.md, oral defence

Key takeaways

- ▣ Gaussian QML needs only the variance equation; with fat tails, use sandwich standard errors
- ▣ Volatility has fast and slow components; macro drivers of the slow one need long samples and honest variance ratios
- ▣ Realised measures make volatility observable with a known error; noise and jumps decide the estimator
- ▣ HAR, HARQ and Realized GARCH beat return-only models one day ahead; evaluate with QLIKE or MSE
- ▣ In many dimensions the correlation target needs shrinkage (DCC-NL); whether the dynamics help must be checked out of sample

Self-assessment

Questions

- ▣ Why are Hessian standard errors too small for GARCH when $\kappa_\eta > 3$?
- ▣ What does the variance ratio of GARCH-MIDAS measure?
- ▣ Why does stable convergence matter for the CLT of RV?
- ▣ Why can a signature plot rise at high frequency?
- ▣ Why is β_{dQ} expected to be negative in HARQ?
- ▣ Which losses keep the ranking under a noisy proxy?

Next:

- ▣ VaR, ES and backtesting: elicibility, scoring and model risk
- ▣ From volatility forecasts to tail forecasts: VaR 1% and ES 2.5%, their scoring functions and backtests

Appendix: the GARCH sandwich

- $\ell_t = -\frac{1}{2}(\ln \sigma_t^2 + \varepsilon_t^2/\sigma_t^2)$, $d_t = \sigma_t^{-2} \partial \sigma_t^2 / \partial \theta$: $s_t = \frac{1}{2}(\eta_t^2 - 1)d_t$ at θ_0
- $B = \mathbb{E} s_t s_t' = \frac{1}{4} \mathbb{E}(\eta_t^2 - 1)^2 \mathbb{E} d_t d_t' = \frac{\kappa_\eta - 1}{4} J$, because η_t is independent of $d_t \in \mathcal{F}_{t-1}$
- $\partial s_t / \partial \theta' = -\frac{1}{2} \eta_t^2 d_t d_t' + \frac{1}{2}(\eta_t^2 - 1) \partial d_t / \partial \theta'$; the second term has mean zero, so $A = \frac{1}{2} J$
- $A^{-1} B A^{-1} = 4J^{-1} \cdot \frac{\kappa_\eta - 1}{4} J \cdot J^{-1} = (\kappa_\eta - 1)J^{-1}$; Hessian only: $A^{-1} = 2J^{-1}$; OPG: $B^{-1} = \frac{4}{\kappa_\eta - 1} J^{-1}$

Appendix: the noise bias of realised variance

- ▣ $\tilde{r}_i = Y_{i/n} - Y_{(i-1)/n} = r_i + u_i - u_{i-1}$, with u i.i.d. $(0, \omega^2)$ independent of X
- ▣ $\sum_i \tilde{r}_i^2 = \sum_i r_i^2 + 2 \sum_i r_i(u_i - u_{i-1}) + \sum_i (u_i - u_{i-1})^2$; the cross term has mean zero, the last has mean $2n\omega^2$
- ▣ $\text{Cov}(\tilde{r}_i, \tilde{r}_{i-1}) = -\omega^2$: first-order negative autocorrelation; the realised kernel adds $2\gamma_1 \approx -2n\omega^2$ back and removes the bias
- ▣ Positively autocorrelated returns (gradual price adjustment, stale prices) give $\gamma_1 > 0$ and a signature plot that rises with the interval; the kernel corrects both signs

Appendix: attenuation and HARQ

- ▣ $IV_{t+1} = c + \phi IV_t + v_{t+1}$, observed $RV_t = IV_t + e_t$, e_t uncorrelated with IV_t and v_{t+1}
- ▣ OLS of RV_{t+1} on RV_t : $\text{plim } \hat{\phi} = \frac{\text{Cov}(IV_{t+1}, IV_t)}{\text{Var}(IV_t) + \text{Var}(e_t)} = \phi\lambda$, $\lambda = \frac{\text{Var}(IV)}{\text{Var}(IV) + \mathbb{E}\text{Var}(e_t)}$
- ▣ The best linear predictor given today's precision uses $\phi\lambda_t$, $\lambda_t = \frac{\text{Var}(IV)}{\text{Var}(IV) + 2IQ_t/n}$: decreasing in IQ_t
- ▣ A first-order expansion in $\sqrt{IQ_t}$ gives $\beta_d + \beta_{dQ}\sqrt{RQ_t}$ with $\beta_{dQ} < 0$: the HARQ specification [Bollerslev, Patton and Quaadvlieg \(2016\)](#)

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